

VESTA: Time-Budgeted Multimodal Agents for Retail Financial Decision Support

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Abstract. In high-volatility emerging markets such as Borsa Istanbul (BIST), retail investors face a low signal-to-noise ratio created by the divergence between unstructured disclosures and rapidly changing charts. Existing decision-support systems process news and technical pictures independently, and they ignore the investor’s temporal bandwidth: the same briefing is delivered whether the user has one minute or ten. This paper presents VESTA, a three-layer agentic system that (i) retrieves portfolio-conditioned textual evidence, (ii) reads candlestick images and the underlying OHLCV numbers, and (iii) spends a declared 1-, 3-, or 10-minute budget by changing the depth of the agentic chain. The contribution is that briefing stack plus an evaluation that treats the user’s minutes as a compute budget and measures leakage, per-tier delivery, and action against buy-and-hold. Text and vision are combined with a standard sigmoid mixer and compared with GMU and tensor fusion. A realized-volatility flag on the same day’s OHLCV is recovered at 100% F1 by a closed-form rule, so the headline task is next-day BIST100 direction. On public BIST100 data (2018-2026), no fusion method significantly outperforms GMU (McNemar $p > 0.46$ on seed 0). Coverage, declared latency, and length-normalized information noise vary across the three time tiers. A paper-trading overlay on seed-0 calls does not beat buy-and-hold in the 2025-2026 test window. Source: <https://github.com/ozge-devops/congress>.

Keywords: Multimodal AI · Agentic RAG · Financial decision support · Borsa Istanbul · Temporal bandwidth · Evaluation protocol

1 Introduction

Retail investors sit between the textual density of the Public Disclosure Platform (KAP) [18] and the visual flux of technical charts. The difficulty is not only volume but modality mismatch: a constructive disclosure can coincide with a failed breakout, and the two streams must be reconciled before a decision. Conventional DSS pipelines keep NLP and computer vision in separate pipelines. A second failure mode is temporal: systems emit the same information density

whether the user has sixty seconds or ten minutes [7, 17, 22]. On a mobile-first retail platform that mismatch adds intrinsic load [23].

Recent LLM trading agents attack related problems from a different angle. FinMem layers memory by information half-life [27]. Neither that line of work nor typical chart-reading trading bots treat the user’s available time as a first-class computational budget, and they are evaluated as trading agents rather than as a retail decision-support interface. VESTA occupies that gap: KAP list text and a 40-bar chart, delivered as a 1-, 3- or 10-minute briefing, with an evaluation that scores leakage, per-tier delivery, and action against buy-and-hold. If the user has about one minute, the system emits a short alert or stays silent. If they have about three minutes, it adds one retrieval step over similar past days. If they have about ten minutes, it expands into a briefing with a macro explanation and a hedge sketch. The fusion rule does not change with the clock; only how many agent steps run and how much text is shown. We measure, for each setting, how often a call is issued, how long the pipeline is allowed to run, how many tokens are delivered, and what fraction of that payload is non-actionable. Decision support is scored on next-day BIST direction, not on a realized-volatility flag that can be read off the same day’s OHLCV table. Whenever a chart encoder is used, a numeric OHLCV baseline is reported beside it [11]. The mixer is a standard sigmoid over concatenated features [2, 12]; we compare it with GMU [2] and tensor fusion [28]. Actionability is a long/short overlay against buy-and-hold, not classification F1 alone. The public chart is bars $t - 40, \dots, t - 1$ and does not contain today’s candle.

2 Related Work

2.1 LLMs and agentic workflows in finance

Financial NLP has moved from sentiment scoring [20] to domain LLMs such as FinGPT [25]. Hallucination on prices and terminology remains a documented failure mode [13], which is why VESTA grounds text in retrieval [15] rather than in unaugmented generation. Tool-using agents [26] and trading-oriented systems with layered memory [27] are the closest architectural relatives. FinMem allocates memory by the staleness of facts; VESTA allocates compute by the staleness of the user’s attention. FinMem-style agents optimize for the trader; VESTA is scored as a briefing engine for a time-constrained retail user.

2.2 Multimodal fusion and chart understanding

GMU learns a multiplicative gate between modalities [2]; Tensor Fusion Networks model unimodal and bimodal interactions via an outer product [28]; MulT uses directional pairwise cross-modal attention [24]. Jiang and Ji add a forget-gate over cross-modal attention maps [12]. VESTA’s mixer is the bimodal special case of this family; we report GMU and TFN as external controls.

On the vision side, Pix2Struct and MatCha pretrain screenshot-to-text models that read plots [14, 16]. In asset pricing, Jiang, Kelly and Xiu show that CNNs

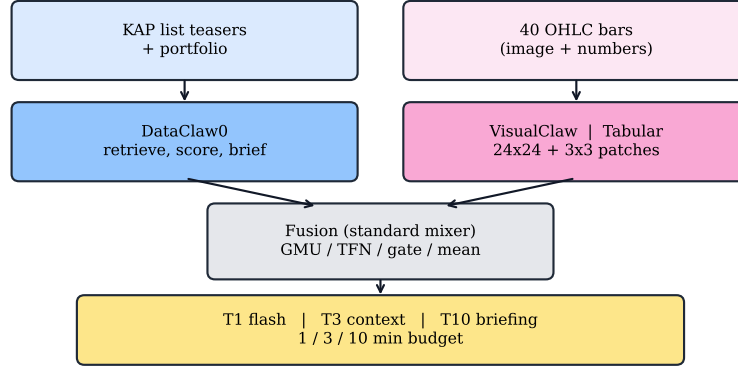


Fig. 1. KAP list text and a 40-bar chart go through a standard mixer; T1, T3 or T10 spends the 1-/3-/10-minute budget.

on price-trend images extract return-predictive patterns [11]. That result motivates VisualClaw and requires a baseline that sees the same series as numbers. Without the tabular arm, a vision score on a label computed from OHLCV is uninterpretable.

BIST100 forecasting from technical indicators is studied by Akturk et al. [1]; their features are numeric, which is the required control whenever a chart encoder is scored.

2.3 Cognitive load and adaptive interfaces

Cognitive load theory requires that presented density match working-memory capacity [22, 23, 17]. Information overload is a long-standing MIS construct [7]. Adaptive GUIs have a documented evaluation discipline: measure whether adaptation helps the task, not only whether the widget changed [9, 8]. Summarization evaluation that ignores output length systematically rewards brevity [21]. On Borsa Istanbul the load is concrete: a KAP list, a candlestick, and a clock between a filing and a retail order. That is the finance instance of the same four literatures. They jointly constrain our metrics (Sect. 4.3) and the 1-/3-/10-minute protocol (Sect. 6).

3 System

VESTA has three layers (Fig. 1): perception by two expert agents, an off-the-shelf fusion block, and a temporal synthesis layer that spends the user’s time budget.

3.1 DataClaw0: portfolio-conditioned retrieval

DataClaw0 is a retrieve-score loop [15, 26] (Fig. 2):

1. Embed the day’s KAP filings (and any news teasers) together with the user portfolio P using M3-Embedding [3].
2. Score each document against the portfolio and the day’s gold / USD/TRY moves. A light sector-to-macro graph says which names are sensitive to those macros. Relevance of document d_i is

$$R(d_i) = \sigma \left(\alpha \cos(v_{d_i}, v_P) + \sum_{j \in \{G, X, T\}} \beta_j \Delta_j \right), \quad (1)$$

where Δ_j is the percentage change of macro-indicator j and σ is the logistic function.

3. Keep the top-ranked filings as the text brief that later enters fusion.
4. Stop after this pass on a one-minute budget (T1). On a three- or ten-minute budget, run one more retrieval hop over historical analogues, then write.

The text dataset is VESTA-Public: one row per ticker-day (37,046 events; 27 BIST names plus XU100; 24 May 2018 to 19 August 2026). A row is kept if the name moved at least 2%, the vol flag fired, RSI was extreme, or a KAP list item was published that calendar day. 19,645 rows have a same-day KAP teaser; 15,212 have a cached public HTML body. Yahoo headlines exist for 41 rows. Table 1 shows four KAP list strings. We do three things with them: (i) a Turkish/English lexicon plus subject priors (dividend and buyback tilt bullish; fine and probe tilt bearish) yields kap_polarity; (ii) the same strings, plus USD/TRY, gold and the session, become an eight-plus-eight feature vector for the mixers; (iii) MiniLM [19] and BGE-M3 embed the concatenated list text. The templated brief a T10 user would read looks like: “2018-06-14 BIMAS.IS. Session return +0.12%. USD/TRY . . . Brief polarity bullish; chart cue none. Pay-ların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri.”

3.2 VisualClaw: candles, pixels, tokens

VisualClaw draws the chart from OHLC bars. Figure 3 is one XU100 window from the test split.

1. Take the 40 sessions strictly before day t (open, high, low, close). Today’s candle is not in the picture.
2. Paint a grayscale candlestick: a one-pixel wick and a rectangular body. Close $\geq$ open is dark (30/255); close $<$ open is light (180/255). Price is min-max scaled to that window.
3. Resize to 24×24 . The vision MLP flattens this to a 576-d vector and classifies with hidden layers 64-32. That is the vision row in Table 4.

Table 1. Example KAP list teasers in VESTA-Public (subject plus short summary from kap.org.tr). Polarity is the silver codebook.

Date	Ticker	KAP list teaser	Polar.
2018-06-14	BIMAS	Payların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri	bullish
2018-06-19	ASELS	BISTECH Pay Piyasası Alım Satım Sistemi Duyurusu. Temettü Ödemesi	bullish
2018-05-25	YKBNK	Özel Durum Açıklaması (Genel). Bankamıza verilen idari para cezasının iptaline ilişkin açılan dava hk.	bearish
2020-08-11	FROTO	Özel Durum Açıklaması (Genel). Rekabet Kurulu Soruşturma Kararı Hakkında	bearish

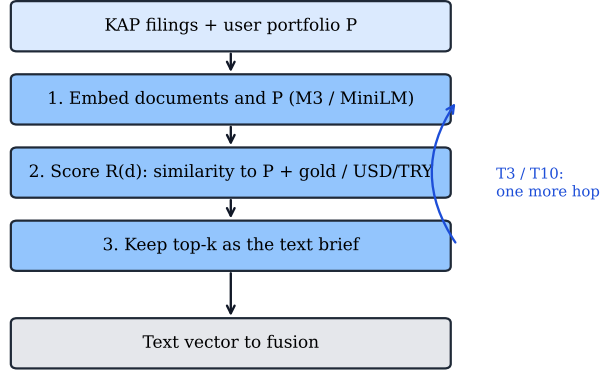


Fig. 2. DataClaw0 retrieve-score loop. The blue hop is a second retrieval, used only on a T3 or T10 budget.

4. For the MulT-style mixer only, cut the same 24×24 image into a 3×3 grid of 8×8 patches (nine visual tokens). The 16-d text brief is split into two tokens. Projections are frozen (seed 0); only the MLP on top is trained.

Two extra probes, not the mixer inputs: a frozen ImageNet ViT-B/16 [6] that resizes a colour chart to 224×224 and uses that model’s own 16×16 patches; and zero-shot DePlot / MatCha, which see a matplotlib RGB screenshot of the same 40 bars (Fig. 3a), not the 24×24 tensor. Tabular OHLCV at day t is always reported beside the picture [11]. A recency-weighted patch position

$$E_{\text{total}} = E_{\text{patch}} + E_{\text{pos}} \cdot (1 + \lambda t/T) \quad (2)$$

is unused in the public tables, which train the 24×24 MLP instead.

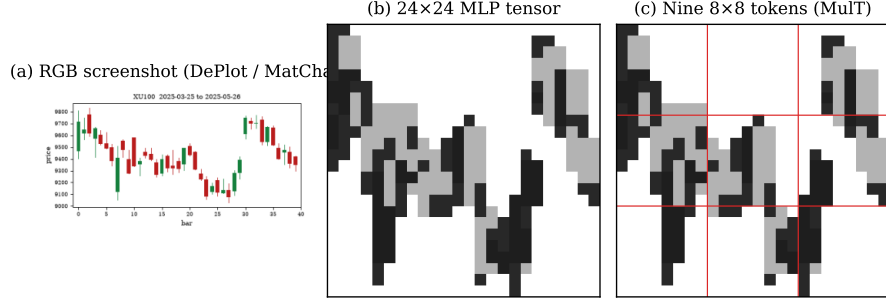


Fig. 3. VisualClaw on one XU100 test window (40 bars before 27 May 2025). (a) RGB screenshot for DePlot / MatCha. (b) 24×24 grayscale tensor the MLP flattens. (c) Nine 8 × 8 tokens for the MulT-style mixer.

3.3 Fusion

The two streams are fused on every run with a standard mixer [2, 12]. Let H_{text} and H_{vision} be the text and vision vectors. A scalar gate

$$g = \sigma(W_g[H_{\text{text}}; H_{\text{vision}}] + b_g), \quad (3)$$

$$H_{\text{fused}} = g \cdot \tanh(H_{\text{text}}) + (1 - g) \cdot \tanh(H_{\text{vision}}). \quad (4)$$

weights the two streams: g close to 1 trusts text, g close to 0 trusts the chart. W_g can be learned. In the public tables we instead mix the two class scores with this gate, with GMU, with tensor fusion, or with a mean, then pass the mix to a small MLP. The same fused score later drives T1 / T3 / T10. All mixers are reported side by side.

3.4 Temporal Orchestration Layer

Given fused confidence $|g - 0.5|$ and a user budget $B \in \{1, 3, 10\}$ minutes, the layer selects a chain:

- T1 (flash). Emit a gated alert only if $|g - 0.5| > \tau$; otherwise abstain. Target latency < 1 s.
- T3 (context). Always emit the fused call plus one retrieval hop over historical analogues.
- T10 (briefing). Expand into a macro explanation and a hedge sketch (e.g. airline exposure versus a gold certificate), at several seconds of wall-clock time.

Depth, not the mixer, is what changes with B . That is the cognitive adaptation: the same fused call is spent as a flash, a short context, or a briefing. Section 5.4 reports coverage, latency, tokens and noise per tier.

Table 2. Proxy accuracy vs next-day direction y_{t+1} (test $n = 308$). Macro/KAP are single-split rules; vision/tabular are mean $\pm$ 95% CI over five seeds.

Proxy	Acc. (%)
Macro flags (USD/TRY, gold, vol, shock)	54.9
KAP polarity (lexicon on public list text)	49.7
Vision 24×24 MLP	52.9 ± 2.3
Tabular OHLCV	50.6 ± 0.0

4 Evaluation Protocol

4.1 Diagnostic: the realized-volatility label

The 2σ realized-vol flag is 20-day vol above the mean $+2\sigma$ of the previous 60 vol observations. It is a closed-form function of the tabular vector at t (100% F1). The public chart does not contain today’s candle, so vision F1 on this label is lag-1 recovery. We report the flag only as a diagnostic.

4.2 Primary task: forward direction

The primary label at close of day t is the next session’s direction: $y_{t+1} = 1$ if the BIST100 simple return r_{t+1} is positive, and $y_{t+1} = 0$ otherwise. The label is unknown at the close of day t . We also report a five-session horizon. Splits are chronological (70/15/15) to block look-ahead [1].

4.3 Metrics

Classification. Accuracy and macro-F1, 95% t -intervals over five seeds. McNemar with continuity correction [5] on seed 0 of one chronological holdout. Table 4 has thirteen rows; McNemar is mixer-only. With that many rows we follow Harvey et al. [10] and do not pick a winner after an uncorrected search.

Sentiment / technical accuracy. Table 2 reports proxy accuracy versus next-day direction. Macro flags and KAP polarity are deterministic rules on one chronological split (no seed CI). Vision and tabular are five-seed means from Table 4. Three KAP codebooks on the 10k slice give Fleiss $\kappa = 0.50$; chart A/B Cohen $\kappa = 0.52$ [4]. The 250-row annotator columns are codebook B.

Information noise. This is the computational cognitive measure. Let N_{raw} be tokens in the unprocessed window (the raw KAP list plus session flags) and N_{del} tokens actually delivered in the T1, T3 or T10 briefing. Let N_{irrel} be delivered tokens that are not in a relevance lexicon for that window (session, USD/TRY, gold, vol, KAP digest). The index $\text{IN}_{\text{old}} = N_{\text{irrel}}/N_{\text{raw}}$ rewards short answers: a 40-token briefing that is 80% filler still looks clean against a 700-token feed [21]. We therefore report

$$\text{IN}_{\text{new}} = N_{\text{irrel}}/N_{\text{del}}, \quad (5)$$

the fraction of the delivered finance briefing that is non-actionable, together with the compression ratio $N_{\text{del}}/N_{\text{raw}}$. Both counts use templated token bags with filler padding. Table 6 is that proxy. It is not a NASA-TLX score.

Decision quality. Classification F1 does not say whether acting on the output helps [8]. We paper-trade a long/short overlay: +1 if the model predicts up, −1 otherwise, with 5 bp round-trip cost, and compare Sharpe, total return and max drawdown to buy-and-hold. Table 7 is that control; Sect. 6 is the human cognitive protocol.

4.4 Public replication corpus

Numbers below use VESTA-Public: XU100.IS, USDTRY and COMEX gold from 24 May 2018 to 19 August 2026 (2,052 index windows after a 40-day lookback; test $n = 308$ from 27 May 2025), 27 BIST names, and 39,890 KAP list filings from kap.org.tr. Tables 3 to 7 are index-level; the 37,046-row event panel also labels constituents. KAP-only polarity: 7,172 / 3,742 / 26,132 (bullish / bearish / neutral); 15,212 rows have a cached public HTML body. Mixers use the 16-d macro+KAP vector; MiniLM and BGE-M3 are unimodal text probes on the same list teasers.

5 Experiments

Unimodal encoders are sklearn MLPs (64-32, Adam, early stopping) on standardized features. Vision sees a 24×24 grayscale candlestick of bars $t - 40, \dots, t - 1$; tabular sees eleven OHLCV/macro features at t , including the ingredients of the leaked vol rule; text sees the sixteen-dimensional macro+KAP vector. Five seeds $\{0, \dots, 4\}$. Training is on CPU. Public BGE-M3 and frozen ViT-B/16 are unimodal probes; they remain unimodal in Table 4.

5.1 Diagnostic: the leaked volatility label

Table 3 and Fig. 4 confirm the closed-form critique. The 2σ rule on the tabular vector scores 100% F1. A tabular MLP that is given those numbers reaches 96.8% accuracy (macro-F1 86.6%; the F1 drop is class imbalance: 9.4% positives). The 24×24 vision encoder, which never sees day t ’s candle, falls to 90.6% accuracy and 58.2% F1.

5.2 Primary task: next-day direction

Table 4 reports the leakage-free task (test prior 52.3% up-days). Text-only (macro+KAP) macro-F1 is 47.3 ± 1.5 against a majority F1 of 34.3; MiniLM 53.0 ± 0.4 / 50.7 ± 1.9 ; public BGE-M3 51.1 ± 1.9 / 50.2 ± 2.1 ; frozen ViT-B/16 50.9 ± 1.4 / 50.3 ± 1.6 . Mixers share 16-d text and the 24×24 image; none beats GMU on seed 0 (min. McNemar $p = 0.46$). Mean fusion is the five-seed

Table 3. Diagnostic task: 2σ realized-vol label, computable from the tabular vector at t . Mean $\pm$ 95% CI over five seeds. Vision sees bars $t - 40, \dots, t - 1$ only.

Method	Accuracy (%)	Macro-F1 (%)
Closed-form OHLCV rule	100.0	100.0
Tabular MLP (same numbers)	96.8 ± 0.0	86.6 ± 0.0
Vision MLP (chart image)	90.6 ± 1.1	58.2 ± 3.0

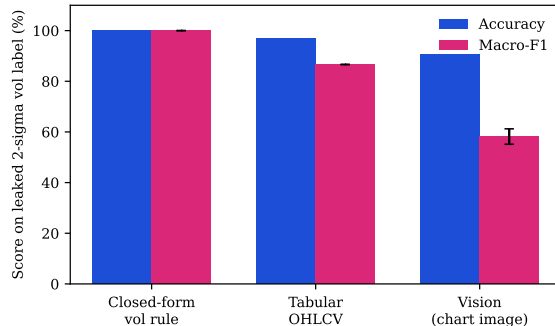


Fig. 4. Diagnostic 2σ -vol label. The closed-form OHLCV rule scores 100% F1; the chart omits day t .

point estimate (53.0 ± 2.2 / 52.4 ± 2.3). Five-day tabular accuracy is 59.1% with collapsed F1 (39.9%).

Replacing the gate with a fixed mean changes five-seed F1 from 51.4 ± 1.8 to 52.4 ± 2.3 (seed-0 McNemar $p = 1.00$).

5.3 Zero-shot Pix2Struct and MatCha

We run zero-shot DePlot (Pix2Struct chart-to-table) [14] and MatCha-ChartQA [16] on CPU matplotlib screenshots of the full test window ($n = 308$), with no fine-tune. Calls are $\text{sign}(\hat{y}_{\text{last}} - \hat{y}_{\text{first}})$. Table 5 also scores the numeric window trend of the input OHLC, which is still not y_{t+1} . DePlot parses 307/308 images and recovers that trend on 77.9% of days; next-day accuracy is 52.9%. MatCha-ChartQA often returns the bar index (parse rate 166/308) and matches the majority class on y_{t+1} .

5.4 Per-tier measurement

Table 6 and Fig. 6 measure the Temporal Orchestration Layer. Latencies 0.9 / 2.1 / 3.4s are declared budgets, not wall-clock. On seed 0, T1 covers 32.5% of days (60.0% among emitted calls; 55.2% after majority fill). T3/T10 always fire at 49.7%. Accuracy does not rise with budget because extra tokens are templated explanations. $\text{IN}_{\text{old}} < 7\%$ for every tier; IN_{new} is 60.3 / 82.1 / 93.6% on filler-padded templates.

Table 4. Next-day BIST100 direction on VESTA-Public (test $n = 308$). Mean $\pm$ 95% CI, five seeds. Rows are ordered by macro-F1. MiniLM, BGE-M3 and ViT are unimodal probes. Mixers: GMU [2], TFN [28], MulT-style [24]. The scalar gate row mixes class scores rather than training W_g in Eqs. (3) and (4).

Configuration	Accuracy (%)	Macro-F1 (%)
Mean fusion	53.0 ± 2.2	52.4 ± 2.3
TFN	52.1 ± 2.1	51.7 ± 2.1
Vision-only (24×24 MLP)	52.9 ± 2.3	51.4 ± 2.6
Scalar gate (score-space)	51.8 ± 1.7	51.4 ± 1.8
Concatenation	52.8 ± 1.8	51.1 ± 2.3
Text-only (MiniLM KAP)	53.0 ± 0.4	50.7 ± 1.9
GMU	50.6 ± 2.4	50.6 ± 2.4
Vision-only (frozen ViT-B/16)	50.9 ± 1.4	50.3 ± 1.6
Text-only (public BGE-M3)	51.1 ± 1.9	50.2 ± 2.1
MulT-style (CPU, one layer)	50.2 ± 3.1	48.4 ± 3.0
Text-only (macro + KAP)	52.1 ± 2.4	47.3 ± 1.5
Tabular OHLCV	50.6 ± 0.0	41.8 ± 0.0
Majority class	52.3 ± 0.0	34.3 ± 0.0

Table 5. Zero-shot CPU screenshot VLMs on the full test window ($n = 308$), with no fine-tune.

Method	Acc. vs y_{t+1}	Macro-F1	Acc. vs window trend
Majority	52.3	34.3	n/a
Numeric window trend	51.6	49.5	100.0
DePlot (Pix2Struct)	52.9	51.9	77.9
MatCha-ChartQA	52.3	46.0	58.4

5.5 Decision quality

Table 7 asks whether acting on the output would have helped. In the May 2025 to August 2026 test window BIST100 buy-and-hold returned +56.3% (Sharpe 1.69). Every seed-0 overlay we tried underperformed that long-only benchmark after 5 bp costs. Vision-only (Sharpe 0.37) and the scalar gate (0.33) were positive but far below BH; GMU and tabular were negative. A bull-market holdout is a hard test for a long/short overlay [10].

6 Cognitive protocol

The cognitive work has two layers. The first is computational and is already in Table 6. A BIST retail user on a mobile desk has about one, three or ten minutes between a KAP drop and an order. That clock is the independent variable. The dependent quantities are coverage (how often the system speaks), declared latency, tokens delivered, and IN_{new} (what fraction of the delivered briefing is non-actionable). The token bags are built from the day’s finance flags: session

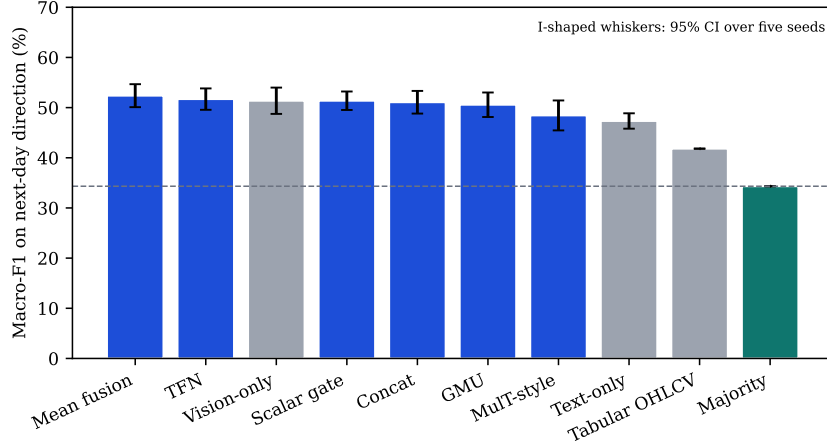


Fig. 5. Same macro-F1 values as Table 4. The I-shaped whiskers are 95% confidence intervals over five seeds (the $\pm$ in the table).

Table 6. Temporal Orchestration Layer on seed 0. Latency is a declared budget, not measured GPU time. Acc. is among emitted calls; T1 overall accuracy with majority fill is 55.2%.

Tier	Cov. (%)	Lat. (s)	Acc. (%)	Tokens	IN _{old}	IN _{new}
T1 flash	32.5	0.9	60.0	6	0.5%	60.3%
T3 context	100.0	2.1	49.7	17	1.9%	82.1%
T10 briefing	100.0	3.4	49.7	52	6.8%	93.6%

return, USD/TRY, gold, realized vol, a KAP digest, and, on T3/T10, a historical analogue and a hedge sketch. Those counts measure load on the payload. They do not measure load on a person [9, 8].

The second layer is a human protocol in **study/**. Twelve retail-account holders, within-subject, see eight sealed snapshots drawn from the public test window of VESTA-Public (two XU100 sessions and six constituents: SASA, SISE, YKBNK, TAVHL, AKBK, PETKM). Each snapshot is a real calendar day. The data that arrive on screen are the session return, the USD/TRY and gold moves, RSI, and the KAP list teasers published that day. The sealed gold is the subsequent week’s realized direction of that name. Participants see one snapshot at a time under a 1-, 3- or 10-minute cap, in one of three information conditions: the raw KAP+OHLC feed, a unimodal text or chart, or the VESTA tier that matches the cap. Order is Latin-squared. Outcomes are the week-direction call, time-to-decision in seconds, six NASA-TLX items on a 0-20 scale (mental demand, physical demand, time pressure, perceived success, effort, frustration), and a 7-point trust item (“I would act on this briefing with real money”). Consent and the bilingual instrument are in **study/**.

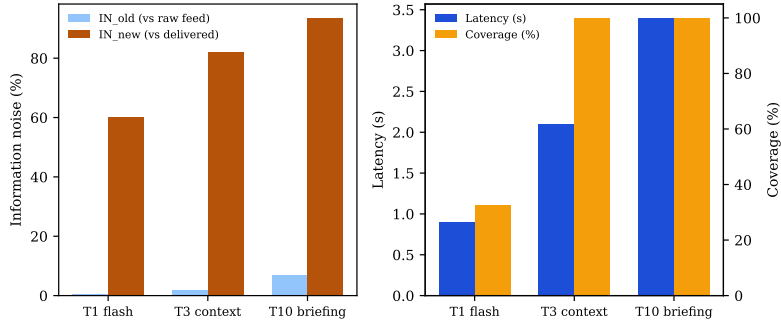


Fig. 6. Left: length-normalized noise of the delivered briefing (IN_{old} vs IN_{new}). Right: declared latency and coverage for T1, T3 and T10.

Table 7. Paper-trading overlay on seed-0 calls, chronological test window, 5 bp costs. BH is long-only BIST100.

Policy	Hit (%)	Sharpe	Total ret.	Max DD
Buy-and-hold	52.3	1.69	+56.3%	14.5%
Vision-only overlay	50.0	0.37	+7.6%	16.4%
Scalar-gate overlay	49.7	0.33	+6.3%	17.1%
Tabular overlay	50.6	-0.11	-6.2%	23.0%
GMU overlay	48.1	-0.74	-21.8%	27.4%

Responses have not been collected. A model dry-run on the sealed gold hits 4/8 name-level weeks for the raw session sign; index mixers are scored only on the two XU100 weeks (text 0/2, gated 1/2). NASA-TLX and trust stay blank.

7 Discussion and limitations

VESTA’s contribution is a time-budgeted briefing for BIST retail users and the evaluation that scores it: time as a budget, leakage made visible, delivery measured per tier, and action compared with buy-and-hold. DataClaw0, VisualClaw and gated delivery are the engineering stack. Next-day accuracy is modest and the overlay loses to buy-and-hold on a bull holdout. The time budget shows up in the tier table: T1 stays silent on most days; T3 and T10 always fire and add templated tokens, so IN_{new} rises with depth.

Public BGE-M3 is run on KAP list teasers; full disclosure-detail pages are not in the package. Frozen ViT-B/16 does not beat the 24×24 MLP. DePlot and MatCha are zero-shot on 308 days. The 12-participant NASA-TLX study has no responses. The public chart uses bars $t - 40, \dots, t - 1$. Information noise and tier latencies are simulated. Table 4 mixers fuse unimodal scores; W_g is not trained.

8 Conclusion

This paper’s contribution is a briefing stack that gives a BIST retail user KAP list text and a 40-bar candlestick under a 1-, 3- or 10-minute budget, together with an evaluation that treats that clock as a compute budget and measures leakage, per-tier delivery, and action against buy-and-hold. Financial LLM agents such as FinMem organise memory by how old a fact is and are scored as traders [27]. Chart-reading and fusion papers typically introduce a new mixer or report a high F1 on a label computed from the same day’s OHLCV. Conventional decision-support systems keep text and charts in separate pipelines and emit the same density whether the user has one minute or ten [7, 22]. That is the gap the sentence above fills. DataClaw0 retrieves KAP list text, VisualClaw reads the candlestick, a standard gate fuses the two streams, and the Temporal Orchestration Layer spends the budget by changing chain depth. The protocol scores next-day BIST100 direction, places a tabular OHLCV baseline beside every chart encoder, measures coverage, declared latency, tokens and length-normalized noise per tier, and paper-trades a long/short overlay against buy-and-hold. VESTA-Public (37,046 events, KAP list teasers, chronological splits) makes those numbers reproducible.

On the public test window ($n = 308$) the benefit is the protocol itself: leakage is visible, tiers are measured, and action is scored against buy-and-hold. A 2σ realized-vol flag is recovered at 100% F1 by a closed-form OHLCV rule, so the headline label is next-day direction. No mixer beats GMU on that task (mean fusion is the point estimate, 53.0 ± 2.2 accuracy / 52.4 ± 2.3 macro-F1; seed-0 McNemar $p > 0.46$). A long/short overlay loses to buy-and-hold in the 2025-2026 window (BH Sharpe 1.69 versus at most 0.37). What the time budget changes is coverage (T1 32.5% on seed 0), declared latency, tokens, and IN_{new} ($60.3 / 82.1 / 93.6\%$ on templated bags). Those four measurements are the literature gap we close: time as a first-class budget, leakage made visible, delivery scored per tier, and action compared with buy-and-hold.

Three items remain open. The computational cognitive measurement is Table 6; the 12-participant NASA-TLX study in **study/** has no responses, so human load and trust are unspecified. There is no three-annotator gold set; polarity is silver codebook output (Fleiss $\kappa = 0.50$ on the 10k slice). Public text is KAP list teasers rather than full disclosure-detail pages, Table 4 mixers fuse unimodal scores rather than a learned W_g , the public chart is bars $t - 40, \dots, t - 1$, IN and tier latencies are simulated, and DePlot / MatCha remain zero-shot on 308 screenshots. Those are the next measurements.

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